Starlight Notes

Every box, macro and code block in starlight.sty ✦ Fall 2026

1 Sets and functions

A function $f\colon A \to B$ is **injective** if $f(a) = f(a')$ implies $a = a'$, and **surjective** if every $b \in B$ has a preimage. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$.

✦ **Definition 1.1** Bijection

A function is a **bijection** if it is both injective and surjective. Equivalently, it has a two-sided inverse f^{-1} with $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$.

✦ **Theorem 1.2** Cantor

For any set A there is no surjection $A \to \mathcal{P}(A)$. In particular $|A| < |\mathcal{P}(A)|$.

Proof. Suppose $f\colon A \to \mathcal{P}(A)$ is onto and let $D = \{a \in A \mid a \notin f(a)\}$. Then $D = f(d)$ for some d, and $d \in D \iff d \notin f(d) = D$, a contradiction. ✦

✦ **Lemma 1.3**

If $g \circ f$ is injective then f is injective.

✦ **Corollary 1.4**

By Theorem 1.2, $\mathbb{R}$ is uncountable, since $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$.

✦ **Example 1.5** A bijection $\mathbb{N} \to \mathbb{Z}$

$f(n) = \frac{n}{2}$ for even n and $f(n) = -\frac{n+1}{2}$ for odd n lists $0, -1, 1, -2, 2, \dots$, hitting every integer exactly once.

✦ **Remark**

Definition 1.1 makes no mention of cardinality; that comes later.

2 Analysis

✦ **Definition 2.1** Limit

$\lim_{x \to a} f(x) = L$ if for every $\varepsilon > 0$ there is $\delta > 0$ with $0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$.

✦ **Proposition 2.2** Integration by parts

$$\int_a^b u\,\mathrm{d}v = \Big[uv\Big]_a^b - \int_a^b v\,\mathrm{d}u, \qquad \frac{\mathrm{d}}{\mathrm{d}x}\left(\int_a^x f(t)\,\mathrm{d}t\right) = f(x).$$

✦ **Problem 1** Series

Evaluate $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ and decide whether $\sum_n \frac{1}{n \log n}$ converges.

Solution

Telescoping, $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$, so the partial sums are $1 - \frac{1}{N+1}$ and the series equals $\boxed{1}$. The second diverges by the integral test: $\int_2^\infty \frac{dx}{x \log x} = \big[\log\log x\big]_2^\infty = \infty$.

Watch out

$\sum a_n$ converging does not follow from $a_n \to 0$; the harmonic series is the standard counterexample.

2.1 Linear algebra

For $A \in \mathbb{R}^{m \times n}$, $\operatorname{rank} A + \dim \ker A = n$, and $\|Ax\|_2 \le \|A\| \|x\|_2$.

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad A^{\mathsf{T}} = A, \quad \operatorname{tr} A = 4, \quad \det A = 3, \quad \langle Ax, x \rangle = 2x_1^2 + 2x_1 x_2 + 2x_2^2 > 0.$$

Tip

To check a matrix is positive definite, all its leading principal minors being positive is quickest by hand.

3 Algorithms

Theorem Master theorem, case 2

If $T(n) = aT(n/b) + \Theta\!\left(n^{\log_b a}\right)$ then $T(n) = \Theta\!\left(n^{\log_b a} \log n\right)$.

Algorithm Binary search

Input: sorted array $A[1..n]$, target x

```
lo ← 1, hi ← n
while lo ≤ hi do
    mid ← ⌊(lo + hi)/2⌋                ▷ never overflows in maths
    if A[mid] = x then return mid
    else if A[mid] < x then lo ← mid + 1
    else hi ← mid − 1
    end if
end while
return NOT FOUND
```

Each iteration halves $hi - lo$, so it runs in $O(\log n)$ time; the C++ version uses `std::lower_bound` for the same thing.

search.cpp

```cpp
#include <vector>
// returns the index of x in a, or -1
int search(const std::vector<int>& a, int x) {
    int lo = 0, hi = static_cast<int>(a.size()) - 1;
    while (lo <= hi) {
        int mid = lo + (hi - lo) / 2;      // no overflow
        if (a[mid] == x) return mid;
        if (a[mid] < x) lo = mid + 1; else hi = mid - 1;
    }
    return -1;
}
```

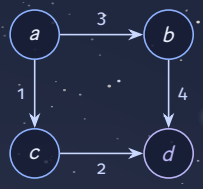
search.py

```python
def search(a: list[int], x: int) -> int:
    """Binary search; returns -1 if x is missing."""
    lo, hi = 0, len(a) - 1
```

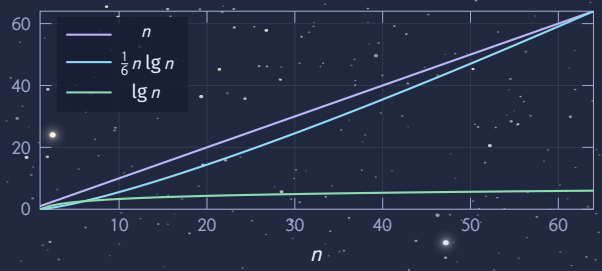
```
    while lo <= hi:
        mid = (lo + hi) // 2
        if a[mid] == x:
            return mid
        lo, hi = (mid + 1, hi) if a[mid] < x else (lo, mid - 1)
    return -1
```

```
$ g++ -std=c++20 -O2 search.cpp -o search && ./search
found 42 at index 17
```

3.1 Graphs and plots



Shortest $a \to d$ path: a, c, d (cost 3).





Structure	Search	Insert
Sorted array	$O(\log n)$	$O(n)$
Balanced BST	$O(\log n)$	$O(\log n)$
Hash table	$O(1)$ avg.	$O(1)$ avg.

3.2 Probability

For independent X, Y: $\mathbb{E}[X+Y] = \mathbb{E}\,X + \mathbb{E}\,Y$, $\operatorname{Var}(X+Y) = \operatorname{Var} X + \operatorname{Var} Y$, and $\mathbb{P}[A \mid B] = \mathbb{P}[A \cap B]/\mathbb{P}[B]$.

✦ Exercise 2

Show that a random permutation of $[n]$ has 1 fixed point in expectation.

1. Write the count as $\sum_i X_i$ with indicators X_i.
2. Use linearity: $\mathbb{E}\,X_i = \frac{1}{n}$.

✦ Summary

- Cantor: $|A| < |\mathcal{P}(A)|$ ✦ telescoping sums ✦ binary search is $O(\log n)$.
- Linearity of expectation needs no independence.